MILITARY SPECIFICATION SHEET

SWITCHES, MULTISTATION, PUSHBUTTON, (ILLUMINATED MATRIX ARRANGEMENT), PLUG-IN MODULE, COMMON TERMINATION SYSTEM, STYLE SM10L1

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-24317.

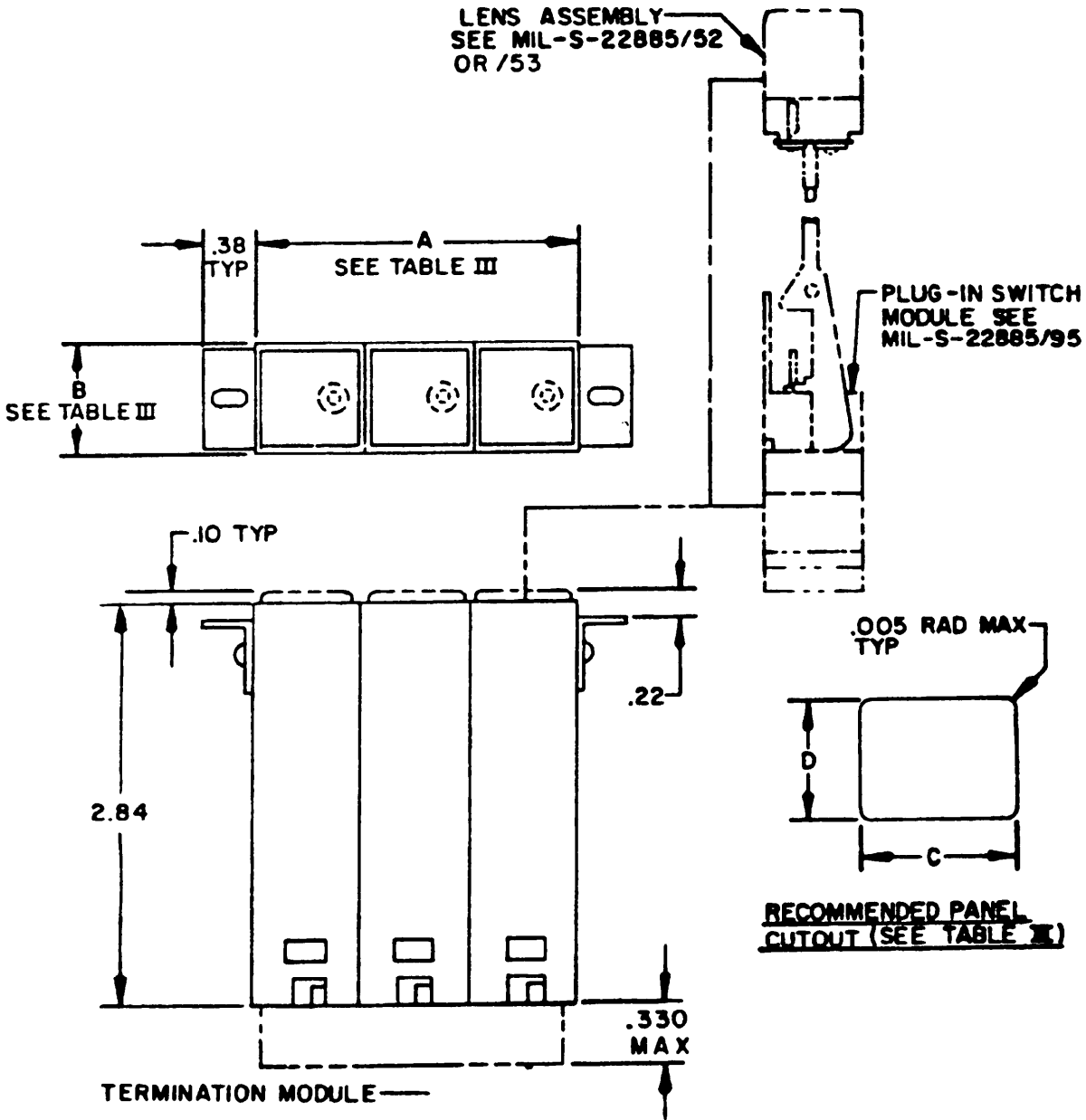
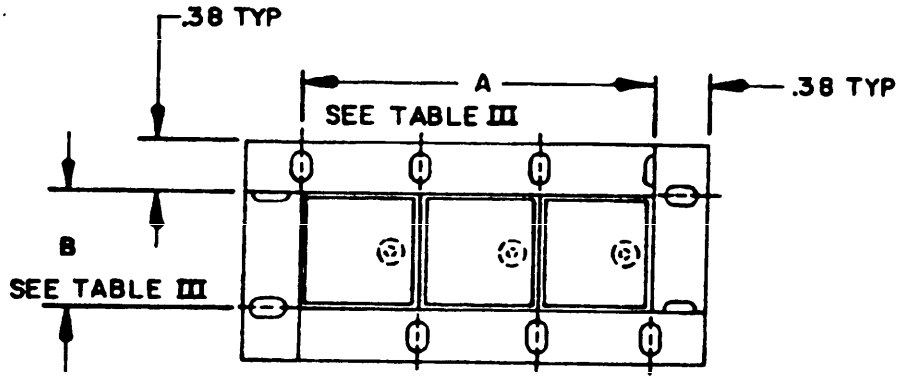


FIGURE 1. Matrix frame assembly, three station.

(B) denotes changes.

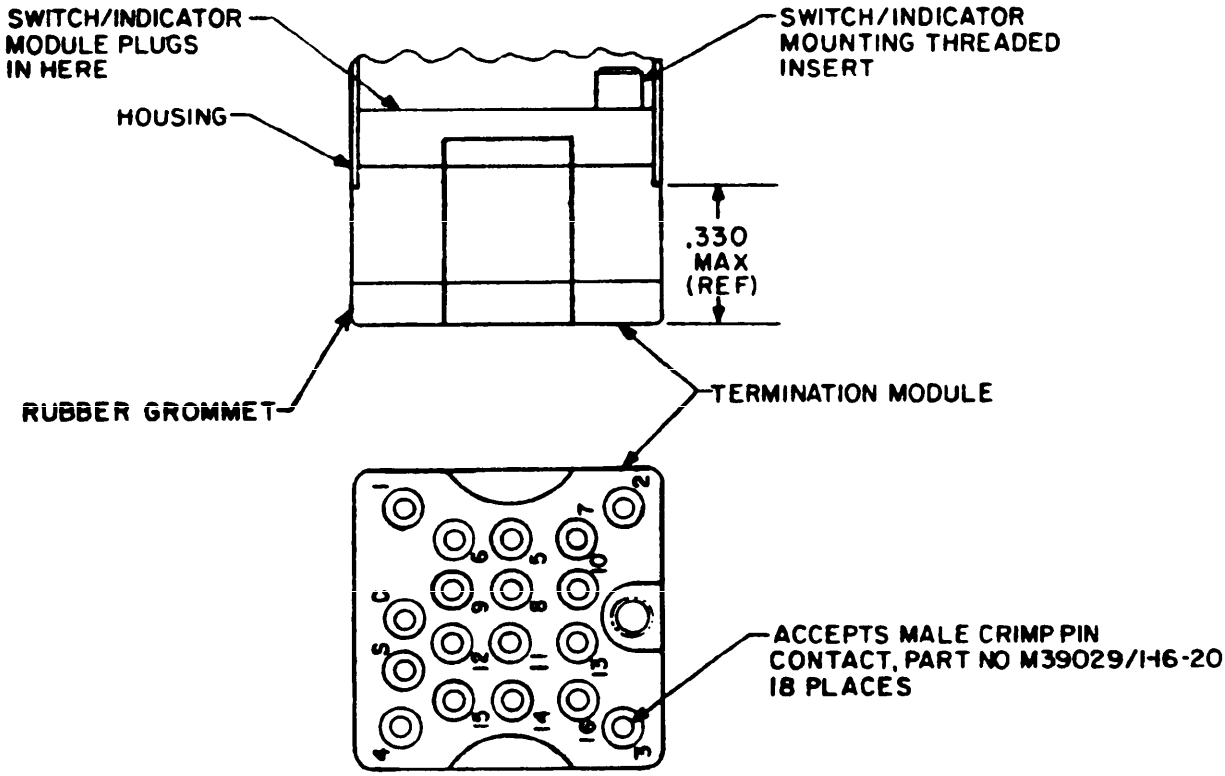
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INCHES	MM
.005	.13
.10	2.5
.22	5.6
.330	8.38
.38	9.7
2.84	72.1

FLANGE ARRANGEMENT FOR RFI SHIELDING



TERMINAL IDENTIFICATION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.015 (.38 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Terminals shall be permanently identified with numbers as shown in terminal arrangement.
5. The design configuration is optional within the envelope dimensions shown.

FIGURE 1. Matrix frame assembly, three station - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, and 3 and table III.

Complete multistation switch assembly consists of:

a. Multistation matrix frame (see table III).

b. Switch module, plug-in type (see MIL-S-22885/95).
Not included in table III part numbers. Order separately.

c. 4 lamps (T 1-3/4 midget flange base incandescent).
Lamps not included. Order separately.

d. Pushbutton lens (see MIL-S-22885/52 or /53).
Not included in table III part numbers. Order separately.

Matrix frame: Includes flanges and termination modules.

Termination module: In accordance with MIL-STD-1549. The female lamp contacts are copper alloy nickel plated .0001 thick per QQ-N-290, class 2. The female switch contacts are copper alloy gold plated per MIL-G-45204, type II, grade C, class 1, over nickel plate .000050 thick. For use with male crimp pin contact, part number M39029/1-16-20.

Frame finish:

Non-RFI shielded - Black anodize per MIL-A-8625.
RFI shielded - Chemical film per MIL-C-5541.

RFI shielding: Requirement and test per MIL-S-22885/95.

Weight (approximate): 1.71 ounces (48.6 grams) per station.

Electrical ratings: See table I.

TABLE I. Electrical ratings.

Load	Sea level	
	28 V dc	115 V ac
	Amperes	Amperes
Resistive -	2.0	2.0
Inductive -	1.0	1.0
Lamp - - - -	0.75	0.75

Operating force: 3.5 ±1.0 pounds.

Operating temperature: -55°C to +85°C.

Thermal shock: During high temperature portion of thermal shock test, all four lamps in lens module shall be energized with full rated voltage.

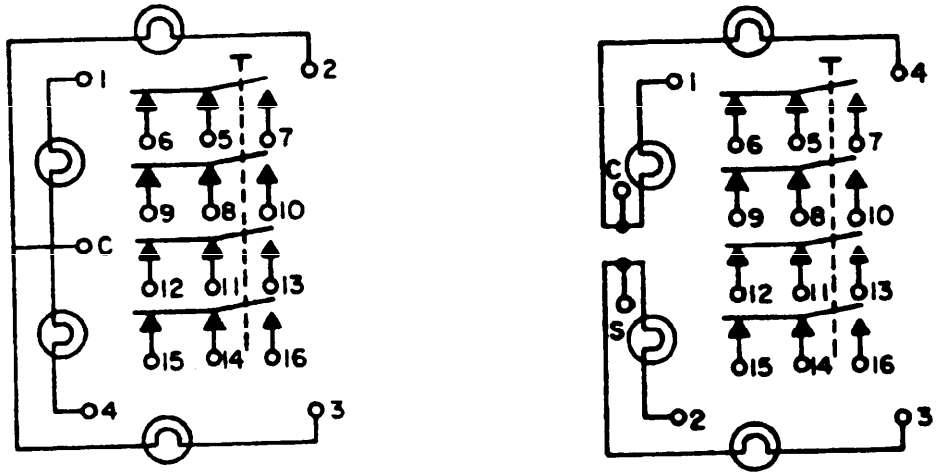
Colors and photometric brightness: In accordance with MIL-S-22885/52 or /53.

Solderability: Not applicable.

Ⓑ Shock: Method I: Tested in accordance with MIL-STD-202, method 213, test condition letter B (75g).
Method II, high impact, applicable to 1 x 1 matrix only, part numbers M24317/10-021 and -031.

Matrix frame part number: M24317/10-(dash number from table III).

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NOTES:
1. Terminals 8 thru 13 are omitted for 2PDT configurations.
2. Terminals 5 thru 16 are omitted for indicator configurations.
3. Terminals 5, 8, 11, and 14 are pole terminals.
4. Terminal S is omitted for common lamp circuit.

FIGURE 2. Lamp and 4PDT switch schematic.

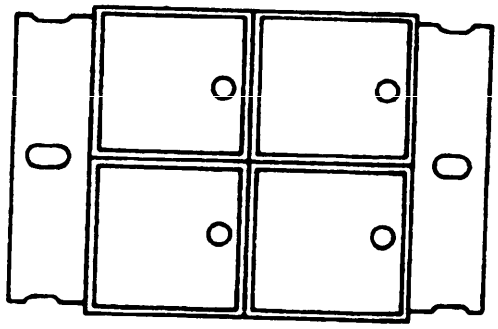


FIGURE 3. Flange arrangement for unshielded switch (vertical flanges shown).

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Qualification inspection: All applicants for qualification approval shall demonstrate that each of their items conforms to all the requirements specified in the applicable documents, singularly and in combination with all other previously qualified items, regardless of manufacturer. When the lens assembly and plug-in switch modules are qualified to MIL-S-22885/52, /53, and /95, qualification and group C inspection of the multistation switch assembly shall be in accordance with table II.

(B) Extent of qualification: The multistation switch/indicator assembly to be used for qualification inspection shall be the 3-station unit (3 x 1 matrix) and the 1-station unit (1 x 1 matrix). Qualification of the units in accordance with table II shall qualify all multistation units up to and including the 20-station unit. The 3-station units shall contain the following items. Location of each item in the matrix will be determined by the testing activity.

Part number	Description	Quantity
M24317/10-023	3 x 1 matrix frame (RFI shielded)	1
M22885/95-003	2 PDT momentary switch module	1
M22885/95-001	4 PDT momentary switch module	1
M22885/95-005	Indicator module	1
M22885/52-	Pushbutton lens	3

The 1-station units shall contain the following items:

Part number	Description	Quantity
M24317/10-031	1 x 1 matrix frame	1
M22885/95-202	4 PDT alternate switch module	1
M22885/52-	Pushbutton lens	1

(B) TABLE II. Qualification inspection (group submission).

Assembly	Qualification inspection table of MIL-S-24317		Extent of approval
	Group	Number of sample units	
3 x 1 matrix (RFI shielded)	I 1/ II 2/ III 3/4/	4 4 from group I 2 from group II	All
1 x 1 matrix	I 1/ II 2/5/	4 4 from group I	

1/ Simultaneity not applicable.
2/ Torque strength of plunger not applicable.
3/ Dielectric withstanding voltage after salt spray not applicable.
4/ Sample units shall be subjected to RFI shielding test of MIL-S-22885/95.
5/ Sample units shall be subjected to method II high impact shock.

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TABLE III. Dash numbers and design and panel cutout dimensions.

Dash number 1/		Number of stations	Dimensions 2/				Number of flanges 3/	RFI shielded
			Matrix ±.015 (.38)		Panel cutout +.030 (.76) -.000			
Common ground lamp circuit	Split ground lamp circuit		A	B	C	D		
-001	-011	1 (1 x 1)	.755 (19.18)	.755 (19.18)	.775 (19.69)	.775 (19.69)	2V	No
-002	-012	2 (2 x 1)	1.510 (38.35)		1.530 (38.86)		2V	No
-003	-013	3 (3 x 1)	2.265 (57.53)		2.285 (58.04)		2V	No
-004	-014	4 (4 x 1)	3.020 (76.71)		3.040 (77.22)		2V	No
-005	-015	5 (5 x 1)	3.775 (95.89)		3.795 (96.39)		2V	No
-006	-016	6 (6 x 1)	4.530 (115.06)		4.550 (115.57)		2V	No
-007	-017	7 (7 x 1)	5.285 (134.24)		5.305 (134.75)		2H	No
-008	-018	8 (8 x 1)	6.040 (153.42)		6.060 (153.92)		2H	No
-009	-019	9 (9 x 1)	6.795 (172.59)		6.815 (173.10)		2H	No
-010	-020	10(10 x 1)	7.550 (191.77)		7.570 (192.28)		2H	No
-021	-031	1 (1 x 1)	.755 (19.18)		.775 (19.69)		2V2H	Yes
-022	-032	2 (2 x 1)	1.510 (38.35)		1.530 (38.86)		2V2H	Yes
-023	-033	3 (3 x 1)	2.265 (57.53)		2.285 (58.04)		2V2H	Yes
-024	-034	4 (4 x 1)	3.020 (76.71)		3.040 (77.22)		2V2H	Yes
-025	-035	5 (5 x 1)	3.775 (95.89)		3.795 (96.39)		2V2H	Yes
-026	-036	6 (6 x 1)	4.530 (115.06)		4.550 (115.57)		2V2H	Yes
-027	-037	7 (7 x 1)	5.285 (134.24)		5.305 (134.75)		2V2H	Yes
-028	-038	8 (8 x 1)	6.040 (153.42)		6.060 (153.92)		2V2H	Yes
-029	-039	9 (9 x 1)	6.795 (172.59)		6.815 (173.10)		2V2H	Yes
-030	-040	10(10 x 1)	7.550 (191.77)		7.570 (192.28)		2V2H	Yes

See footnotes at end of table.

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TABLE III. Dash numbers and design and panel cutout dimensions - Continued.

Dash number 1/		Number of stations	Dimensions 2/				Number of flanges 3/	RFI shielded
			Matrix ±.015 (.38)		Panel cutout +.030 (.76) -.000			
Common ground lamp circuit	Split ground lamp circuit		A	B	C	D		
-041	-051	2 (1 x 2)	.755 (19.18)	1.510 (38.35)	.775 (19.69)	1.530 (38.86)	2V	No
-042	-052	4 (2 x 2)	1.510 (38.35)		1.530 (38.86)		2V	No
-043	-053	6 (3 x 2)	2.265 (57.53)		2.285 (58.04)		2V	No
-044	-054	8 (4 x 2)	3.020 (76.71)		3.040 (77.22)		2V	No
-045	-055	10(5 x 2)	3.775 (95.89)		3.795 (96.39)		2V	No
-046	-056	12(6 x 2)	4.530 (115.06)		4.550 (115.57)		2V	No
-047	-057	14(7 x 2)	5.285 (134.24)		5.305 (134.75)		2H	No
-048	-058	16(8 x 2)	6.040 (153.42)		6.060 (153.92)		2H	No
-049	-059	18(9 x 2)	6.795 (172.59)		6.815 (173.10)		2H	No
-050	-060	20(10 x 2)	7.550 (191.77)		7.570 (192.28)		2H	No
-061	-071	2 (1 x 2)	.755 (19.18)		.775 (19.69)		2V2H	Yes
-062	-072	4 (2 x 2)	1.510 (38.35)		1.530 (38.86)		2V2H	Yes
-063	-073	6 (3 x 2)	2.265 (57.53)		2.285 (58.04)		2V2H	Yes
-064	-074	8 (4 x 2)	3.020 (76.71)		3.040 (77.22)		2V2H	Yes
-065	-075	10(5 x 2)	3.775 (95.89)		3.795 (96.39)		2V2H	Yes
-066	-076	12(6 x 2)	4.530 (115.06)		4.550 (115.57)		2V2H	Yes
-067	-077	14(7 x 2)	5.285 (134.24)		5.305 (134.75)		2V2H	Yes
-068	-078	16(8 x 2)	6.040 (153.42)		6.060 (153.92)		2V2H	Yes
-069	-079	18(9 x 2)	6.795 (172.59)		6.815 (173.10)		2V2H	Yes
-070	-080	20(10 x 2)	7.550 (191.77)		7.570 (192.28)		2V2H	Yes

1/ Dash numbers shown are for matrix frames with flanges and termination modules and do not include switch/indicator or lens assemblies. See MIL-S-22885/95.

2/ Dimensions are in inches. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm. See figure 1.

3/ See figure 3 (2V = two vertical flanges; 2H = two horizontal flanges).

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